Solid State Physics — Week 9 Assignment Questions & Answers

Question 81

Find an expression for density of states of free electrons D(E) in a one-dimensional solid having dimension a.

(a) $\frac{a}{2\pi}\sqrt{\frac{2m}{\hbar^2}}E^{\frac{-1}{2}}$

(b) $\frac{a}{2\pi}\sqrt{\frac{2m}{\hbar^2}}E^{\frac{1}{2}}$

(c) $\frac{a}{2\pi}\sqrt{\frac{2m}{\hbar^2}}E^{\frac{3}{2}}$

(d) $\frac{a}{2\pi}\sqrt{\frac{2m}{\hbar^2}}E^{\frac{5}{2}}$

Accepted answer: (a)

Question 82

Given that the effective mass of the electron is 0.2 times its rest mass, calculate its momentum when its energy$0.02\ eV$. Consider the electron at the bottom of the conduction band and the rest mass of the electron$(m_0 = 9.1 \times 10^{-31}\ kg)$. Planck constant is$(h = 6.62 \times 10^{-34} J - s)$

(a) $1.74 \times 10^{-23}\ kg.m/s$

(b) $3.41 \times 10^{-26}\ kg.m/s$

(c) $5.32 \times 10^{-29}\ kg.m/s$

(d) $7.23 \times 10^{-32}\ kg.m/s$

Accepted answer: (b)

Question 83

The energy wave vector dispersion relation for a one dimensional crystal of lattice constant 'a' is given by $E(k) = E_0 - \alpha - 2\beta\cos(ka)$ where $E_0, \alpha \; and \; \beta$ are all constant. Find the value of k at which the velocity (group velocity) of electron is a maximum, and also find out the maximum velocity.

(a) $k = \frac{n\pi}{2a}, v_{max} = \frac{2\beta a}{\hbar}$

(b) $k = \frac{n\pi}{a}, v_{max} = \frac{2\beta}{\hbar}$

(c) $k = \frac{(2n+1)\pi}{a}, v_{max} = \frac{2a}{\hbar}$

(d) $k = \pi, v_{max} = \hbar a$

Accepted answer: (a)

Question 84

The energy (E) and wave vector (k) dispersion relation for an electron in a one dimensional crystal of lattice constant 'a' is given by $E(k) = E_0 - \alpha - 2\beta\cos(ka)$ where $E_0, \alpha \; and \; \beta$ are all constant. Find the difference in energy i.e., energy gap between the top and the bottom of the energy band.

(a) β

(b) 2β

(c) 3β

(d) 4β

Accepted answer: (d)

Question 85

The energy (E) and wave vector (k) dispersion relation for an electron in a one dimensional crystal of lattice constant 'a' is given by $E(k) = E_0 - \alpha - 2\beta\cos(ka)$ where E_0, α and β are all constant. Find the effective mass m^* of the electron at the bottom and the top of the energy gap.

(a) $\frac{\hbar^2}{2\beta a^2}, \frac{-\hbar^2}{2\beta a^2}$

(b) $\frac{\hbar^2}{2a^2}, \frac{-\hbar^2}{2a^2}$

(c) $\frac{\hbar^2}{\beta}, \frac{-\hbar^2}{\beta}$

(d) $\frac{\hbar^2}{2\beta}, \frac{-\hbar^2}{2\beta}$

Accepted answer: (a)

Question 86

The energy of the electron near the valence band edge of a crystal is given by$(E = -Ak^2)$, where$(A = 10^{-39}\, J - m^2)$. An electron with wave vector $\vec{k} = 10^{10}\hat{\imath}\, m^{-1}$ is removed from the completely filled valance band. Find the effective mass and momentum of the hole. Planck constant is$(h = 6.62 \times 10^{-34} J - s)$

(a) $m_h^* = 5.5 \times 10^{-30}\, kg,\ \vec{P}_h = -1.05 \times 10^{-24}\hat{\imath}\, (J.s/m)$

(b) $m_h^* = -5.5 \times 10^{-30}\, kg,\ \vec{P}_h = 1.05 \times 10^{-24}\hat{\imath}\ (J.s/m)$

(c) $m_h^* = 3.5 \times 10^{-20}\, kg,\ \vec{P}_h = -2.05 \times 10^{-29}\, \hat{\imath}\, (J.s/m)$

(d) $m_h^* = 4.5 \times 10^{-35}\, kg,\ \vec{P}_h = -3.09 \times 10^{-35}\, \hat{\imath}\, (J.s/m)$

Accepted answer: (a)

Question 87

Compute the ratio between the kinetic energies of a free electron in two dimensional square lattices when $(a)\ k_x = k_y = \frac{\pi}{a}$ and when $(b)\ k_x = 0\ and\ k_y = \frac{\pi}{a}$

(a) 2

(b) 4

(c) 6

(d) 8

Accepted answer: (a)

Question 88

Using Kroning-Penny model $\left[\frac{P\sin(\alpha a)}{\alpha a} + \cos(\alpha a) = \cos(ka)\right]$ calculate the energy of the electron for the lowest energy band at k = 0. Consider $P \ll 1$. Where 'a' is the interatomic separation and $\left(\alpha = \sqrt{\frac{2mE}{\hbar^2}}\right)$.

(a) $\frac{Ph^2}{4\pi^2 ma^2}$

(b) $\frac{Ph^2}{\pi^2 ma^2}$

(c) $\frac{Ph^2}{2\pi^2 ma^2}$

(d) $\frac{Ph^2}{3\pi^2 ma^2}$

Accepted answer: (a)

Question 89

Estimate the band gap of an insulator if it absorbs electromagnetic radiation of the wavelengths below the 3000Å. Consider Planck constant($h = 6.62 \times 10^{-34} J - s$).

(a) 3.13 eV

(b) 4.14 eV

(c) 5.27 eV

(d) 5.86 eV

Accepted answer: (b)

Question 90

Calculate the ratio $\frac{k_F}{k_m}$ for a FCC metal having valency one. Where k_F the radius of the sphere of Fermi is surface and k_m is the shortest distance of the hexagonal face from the centre of the first Brillouin zone of FCC crystal.

(a) 0.701

(b) 0.802

(c) 0.902

(d) 1.302

Accepted answer: (c)